

Nidhi Singh

M.Sc. Economics, Dept. of Economic Sciences
Indian Institute of Technology Kanpur

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2025–Present	M.Sc. Economics	Indian Institute of Technology Kanpur	–
2023	B.Sc. (H) Mathematics	Daulat Ram College, University of Delhi	8.37/10
2020	Class XII (CBSE)	Ivey International School	88.0%
2018	Class X (CBSE)	Ivey International School	87.4%

Scholastic Achievements

- Participated in research paper writing competition, **Bhartiya Shikshan Mandal–Yuva Aayam (Delhi Region)** 2022
- Participated in **Ganit ka Mahasangam**, organized by Adarsh Vikash Sanstha (Uttar Pradesh) 2019

Key Projects

Financial Risk Analysis: Loan Default Prediction with Ensemble Learning (*Self Project*) May 2025–Aug 2025

- Utilized **LendingClub** loan dataset (10,000+ applications) with borrower demographics, financial history, and repayment status
- Developed ensemble model combining **Random Forest, Gradient Boosting, and XGBoost** for robust prediction
- Implemented preprocessing pipeline: missing value imputation, categorical encoding, and feature scaling
- Optimized hyperparameters with **GridSearchCV** and applied soft voting strategy
- Achieved **0.87 ROC-AUC**, **84% precision**, and **81% recall**, improving classification of high-risk borrowers by **19%**

Marketing Analytics: Advertising ROI Prediction Model(*Self Project*) jun 2025 - jul 2025

- Designed and trained models on a **synthetic advertising dataset (500+ campaigns)** with ad spend, duration, channel, and market growth rate
- Developed ensemble regression models: **Random Forest, Gradient Boosting, and Stacking Regressor**
- Implemented preprocessing pipeline with **feature scaling, categorical encoding, and data splitting**
- Tuned hyperparameters using **cross-validation** to improve generalization
- Achieved **$R^2 = 0.91$** , reducing prediction error (**MAE = 2.8**, **MSE = 17.8**) compared to baseline regression

Time Series Forecasting: Electricity Consumption Prediction(*Self Project*) mar 2025 - jul 2025

- Analyzed 200 days of electricity consumption, temperature, and holiday data using Python (Pandas, NumPy, Matplotlib)
- Implemented **ARIMA** and **Facebook Prophet** models, achieving RMSE of 1.94 kWh and MAPE of 2.8%
- Performed feature importance analysis revealing temperature as the highest predictor of consumption patterns
- Built forecasting dashboard for visualizing daily and weekly consumption trends, improving prediction accuracy by 18% over baseline

Macroeconomics: Inflation Forecasting using Time-Series Models (*Self Project*) Jan 2025 – Apr 2025

- Collected monthly **CPI and macroeconomic indicators (2010–2024)** from RBI and World Bank databases
- Implemented **ARIMAX, VAR, and LSTM** models to forecast India's monthly inflation
- Conducted stationarity tests (ADF), feature scaling, and multicollinearity checks
- VAR analysis revealed **oil prices and exchange rate** as key drivers of inflation dynamics
- Achieved **MAPE = 3.2%** with LSTM, improving short-term forecast accuracy by 10% over ARIMAX

Positions of Responsibility

Volunteer — Women Development Cell, Daulat Ram College, DU 2022 – 2023

- Participated in international meetings on women's development and alumni networking.
- Organized gender awareness campaigns in campus.

Volunteer — Sponsorship Team, Student Union, Daulat Ram College, DU Jan 2023 – Apr 2023

- Organized Miss DRC 2023 event in North Campus.
- Secured sponsorships for the annual fest Manjari 2023.

Technical Skills

Programming: Python, R, $\text{\LaTeX}$, Mathematica

ML/Stats Tools: Scikit-learn, Pandas, NumPy, Statsmodels, Ensemble Methods

Relevant Courses

**currently enrolled course*

Probability and Statistics	Vector Algebra
Real Analysis and Multivariable Calculus	Linear Algebra
Modern Algebra	Partial Differential Equations
Ordinary Differential Equations	Numerical Analysis
Linear Programming and Applications	Introduction to Machine Learning*

Extra Curricular Activities

- Table Tennis** Finalist in School annual sports event 2015
- Represented school **Kho Kho** team in Moon Olympics Agra 2016